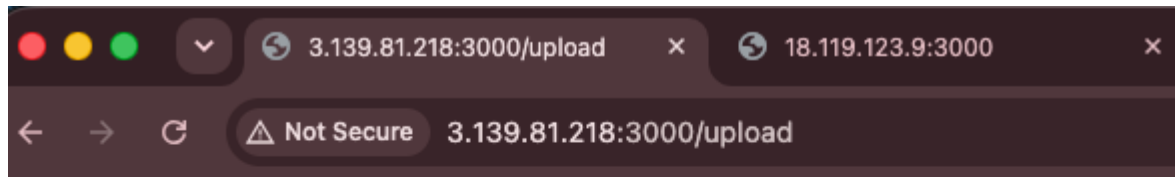


## 1.local folder

```
root@e4d554fa1453:/ITMD544/project/lab3# hostname
e4d554fa1453
root@e4d554fa1453:/ITMD544/project/lab3# ls
create_app_stack.sh      uploader-lt-data.json    viewer-app
create_app_stack.sh.bak  uploader-policy.json     viewer-lt-data.json
delete_app_stack.sh      uploader_instance_id.txt viewer-policy.json
trust-policy.json        user-data-uploader.sh    viewer_instance_id.txt
uploader-app             user-data-viewer.sh
root@e4d554fa1453:/ITMD544/project/lab3# cd uploader-app/
root@e4d554fa1453:/ITMD544/project/lab3/uploader-app# ls
app.js  package.json
root@e4d554fa1453:/ITMD544/project/lab3/uploader-app# cd ..
root@e4d554fa1453:/ITMD544/project/lab3# cd viewer-app/
root@e4d554fa1453:/ITMD544/project/lab3/viewer-app# ls -llrt
total 8
-rw-r--r-- 1 501 dialout 156 Oct  3 03:34 package.json
-rw-r--r-- 1 501 dialout 961 Oct  3 03:49 app.js
root@e4d554fa1453:/ITMD544/project/lab3/viewer-app#
```

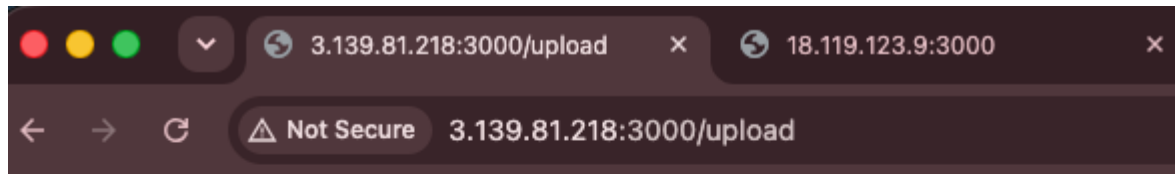
## 2.Uploader page



Upload successful.

[Upload another](#)

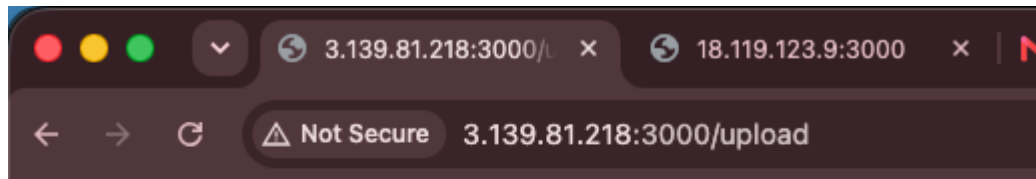
3. >1m



Upload failed: file exceeds the 1MB limit.

[Try again](#)

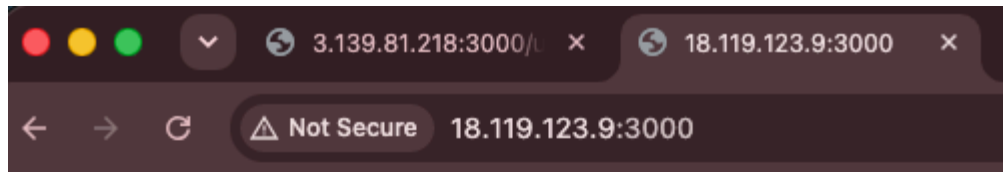
4.Only Text



Upload failed: only .txt files are allowed.

[Try again](#)

5.Viewer page



# Viewer

## CS550 PA1 – Simple P2P File Sharing

### 1. Requirements

Python 3 and three Ubuntu VMs. No extra Python packages are needed. Run commands from your project folder, for example:

```
cd ~/CS550/p2p
```

If you moved the project, use ~/data/CS550/p2p instead.

### 2. Configuration

Check config.json on each peer before starting:

VM1: peer\_a, host 10.0.2.5, port 9001, shared\_dir peer\_a/share

VM2: peer\_b, host 10.0.2.15, port 9002, shared\_dir peer\_b/share

Both peers use index host 10.0.2.6 and index port 9000.

Use listen\_host 0.0.0.0. Update the IP addresses if they changed.

### 3. Start the services

Start VM3 first:

```
python3 server.py --host 0.0.0.0 --port 9000 --replication-factor 3
```

Start Peer A on VM1:

```
python3 peer_server.py --config config.json
```

Start Peer B on VM2:

```
python3 peer_server.py --config config.json
```

Keep all three terminals open. Use new terminals for the following.

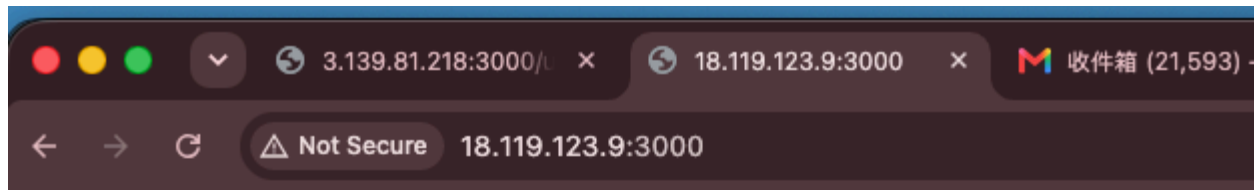
### 4. Register files

Put files in each peer's shared directory. On BOTH VM1 and VM2,

```
python3 register.py --config config.json --no-replicate
```

Repeat registration on both peers after restarting the index server.

6.Refresh



# Viewer

## CS550 P2P – STARTUP AND RESTART SEQUENCE

Applies to the source in p2p\_full.tar.gz reviewed September 28, 2026.

### ORDER

1. VM3: start index.
  2. VM1: start Peer A.
  3. VM2: start Peer B.
  4. VM1 and VM2: register files.
  5. Search/download; then run benchmarks if needed.
- Leave the three service terminals open. Run client commands in extra terminals.

### A. FIRST CHECK EACH VM

```
hostname
hostname -I
```

### Recorded addresses:

VM3 / index: 10.0.2.6:9000

VM1 / peer\_a: 10.0.2.5:9001

VM2 / peer\_b: 10.0.2.15:9002

If addresses changed, update each peer's config.json and suite.json as needed.

Enter the actual current project directory ON EACH VM:

```
cd /home/vboxuser/data/CS550/p2p
```

If your files are still in the earlier directory instead, use:

```
cd /home/vboxuser/CS550/p2p
```

Choose the directory holding your current code and config.json. Do not run both copies. Commands below assume you are already in the correct project directory.

```
pwd
ls server.py peer_server.py register.py config.json
```

### B. VERIFY PEER CONFIGURATION

VM1 config.json:

```
{
  "index": {"host": "10.0.2.6", "port": 9000},
  "peer": {
    "id": "peer a", "host": "10.0.2.5", "listen host": "0.0.0.0".
```

7.awc ec2 command

```
root@e4d554fa1453:~#  
root@e4d554fa1453:~# aws ec2 describe-launch-templates  
{  
  "LaunchTemplates": [  
    {  
      "LaunchTemplateId": "lt-0456d9df066dd776e",  
      "LaunchTemplateName": "itmo-444-544-uploader-lt",  
      "CreateTime": "2026-10-03T20:21:49+00:00",  
      "CreatedBy": "arn:aws:iam::786589010080:user/lab-user",  
      "DefaultVersionNumber": 1,  
      "LatestVersionNumber": 1,  
      "Operator": {  
        "Managed": false  
      }  
    },  
    {  
      "LaunchTemplateId": "lt-030d90d79eb9dc6c1",  
      "LaunchTemplateName": "itmo-444-544-viewer-lt",  
      "CreateTime": "2026-10-03T20:21:53+00:00",  
      "CreatedBy": "arn:aws:iam::786589010080:user/lab-user",  
      "DefaultVersionNumber": 1,  
      "LatestVersionNumber": 1,  
      "Operator": {  
        "Managed": false  
      }  
    }  
  ]  
}
```

root@e4d554fa1453:~#



```
root@e4d554fa1453:~# aws ec2 describe-launch-template-versions --launch-template-name itmo-444-544-uploader-lt
{
  "LaunchTemplateVersions": [
    {
      "LaunchTemplateId": "lt-0456d9df066dd776e",
      "LaunchTemplateName": "itmo-444-544-uploader-lt",
      "VersionNumber": 1,
      "CreateTime": "2026-10-03T20:21:49+00:00",
      "CreatedBy": "arn:aws:iam::786589010080:user/lab-user",
      "DefaultVersion": true,
      "LaunchTemplateData": {
        "IamInstanceProfile": {
          "Name": "itmo-444-544-uploader-profile"
        },
        "ImageId": "ami-0a1ff75d5f5ce322c",
        "InstanceType": "t3.micro",
        "KeyName": "ITMO-444-544-ITMO-444-544-lab-keypair",
        "UserData": "IyEvYmVudl2Jhc2gKZG5mIHVhZGF0ZSAteQpkbmYgaW5zdGFsbCAtaSBub2RlanMKbWtkaXIgLXAgaL29wdC9hcHAKZWNoYiAiWlT5dWmZUWdaWgh3Y21WemN5QTlJSEp5Y1hWcGNtVw9KM1Y0Y0hKbGMzTW5LVHNLWTI5dWmZUWd1WFZzZEdweU1EMGdjvBvZ4ZFseVpT225iWFZzZEdweUp5azdBbU5Y2m5OME1Ic2dVek5EYkdsbGJGUxNXJRkIXZEU5aWFTVmpkRU5Y2lxcagJtUWdmU0E5SUhKbGNYVnBjbVbVvSjBcAGQzTXRjM1JyTDJOC2FVXvNvKqzF6T1XjceE93b0tZmJ11YzNRZ21LYqndJRD8Nw1hod2NVnpjEwdw3dWamIyNXpkQ0JOUz0Zl1JrbE1SVj1UU1ZwRk1EMGdU0FxsURF0d1qUWdLaUf4TURJME95QXZMeUf4VFVJZ0tERXNNRFE0TERVM05pQm1lWFJ3sY31rS0ntTnzIbkk4wSUhWd2JH0Wha0QE5SUcxMwJlUmxjajWg3Q21BZ2MzUnZjbuZUw1RvZ2JYVnNKR1Z5T0x0bG3XOX11Vkw4YjNKAfOyVW9LU3dLUNCc2FXMXBkSE02SUhzZ1ptbHNAvK5wZW1VnK1FMUJXRj1HU1V4R1gxTKpXa1VnZ1N3S01DQm1hV3hsUm1sc2RHNv1PaUFvY21WemXxQm1hV3hsTENCa1pa2dQVDRNzXdxvZ01DQWdZmJ11YzNRZ2FYTlV1SFFnUFNCbWFXeGxMbTFwY1dlWmGVYQmxJRDA5UFNBbmRHVjRkQz13YkdGcGjPy2dKaV1nWm1sc1pTNXZjbWxuYVc1aGJHNHw1V1VZEc5TWIz2Gv1ja0F5vZ1Vjb0tEtnDxbvJ6vJ1ZEnZjcnRZ2V3b2dJQ0FnSUNBCeP9Y3bjbTrnWtJJb2J1tvjJNJRvZ5Y205eUtdZ7FBUa3haWDFSWZGOUJURXhQVjBWRUp5a3BPd29nSUNBZ2ZRB2dJQ0FnWtJ3b2JuvNn1Q3dnZehKMVpTazdDaUfNz1N3S2ZTazdDZ3BqYj1IemDRQnpNeU5E5Uc1bGR5Q1RNM5e3YVdWdWRDAdDjSEp5wjJsdmJqb2djSEp2WTJWemN5Nwx1b11UVvZkvFgxSkSZSMGxQVGLCOUtu0tZmJ11YzNRZ21FsVKTMRFTVSUQWdZ2N1SNZM1ZV6Y3k1bG3uXVVRbFEZUzBWVvgwNUUJVU3Q210dmJuTjBJRk1QVwXrZ1BTQndjbt1qWlH0ekxtVnVkaTVRVDFKVU1IeDhJRE13TURBN0NncGhjSEF1WjJWMEtDY3ZKeXdnS0hKbGNTd2djbvZ6S1NBOVBpQjdDaUFNfY21WekuTmxibVfVwYFvZ01DQWdQR2d4UGxWd2JH0WhaR1Z5UEM5b01UNETJQ0FfNVRUR4bWiZSnRJRZfSEdod1PEMG1VRTlUVkNjZ11XTjBhVz11UFNj4bWjRyQnNiMkZrSW1CbGjTtjB1WEJsUFNKdGRXEDbHwEJoY25Rd1ptOX1iUzFrVhWsaElqNETJQ0FnSUNBZ1BHBHVjSFYwSUhSNWNHVT1JbVpwYkdVaU1HNHw1V1U5SW5SNGRHWnBiR1VpSUDGalKyVndkRDBpTG5SNGRDSWdjvZ4ZDFseVpXUSTDaUfN5UNBZ01EeHdQaz11YkhrZ0xuUjRkQ0ZtYVd4bGn5d2diV0Y0SURGT1fQd3Z1RDLRSUNBZ01DQWdQR0x0cZEHsDmJpQjB1WEJsUFNKemRXSNRhWFFpUGxWd2JH0WhaRhD2W5WGRHOXVQZ29nSUNBZ1BDOW1iM0p0UGdvZ01HQXBPd3A5S1RzS0NtRndjQzV3YjNOMeDY3ZKWEJzYjJGa0p5d2dLESp5Y1N3Z2NtVnpLU0E5UG1CN0NpQWdKwE3ZyJ1JGa0xuTnBibWRZ1NbnmR1adnLBu0E5UG1CN0NpQWdJ03dJwWm1Bb1pYsN1LU0I3Q21BZ01DQWdJR2xtSUNobGnuSXVZmJ1r1WNB0VBUMGdKMHHKVFvSvVgWwKpURVZmVTBsYVJTY3BJSHNLSUNBZ01DQWdJQ0J5W1hSMWntNgdjbvZ6TG5OMF1YUjFjEwCwTVRNCExuTmxibVfVSnP4d1BsVndiRz1oWKNcbV1XbHNAV1E2SUDacGJHVWdaWghqW1dwa2N5QjbhR1VnTVUxQ01HeHB1Z1wWtGp3dmENENDhZ0UjYy21WbVBTsXZJaJjVYV25zrZ11XZGhvhZq4TDJfK0p5azdBaUfN5UNBZ01IEMtJQ0FnSUNBZ2FXWwdLR1Z5Y2k1dFpYtNpZV2RsSUQwOVBTQW5UMDVNV1Y5VvdGUmZRVXhNVDFKR1JDY3BJSHNLSUNBZ01DQWdJQ0J5W1hSMWntNgdjbvZ6TG5OMF1YUjFjEwCwTVRVCExuTmxibVfVSnP4d1BsVndiRz1oWKNcbV1XbHNAV1E2SUC5dWJ1a2dMb1I0ZENCbWFXeGxjeUJoY21VZ11XhNiM2RswKMOEwzQStQR0VnYhKbFpqMGLMeUkrVkhKNU1HRm5ZV2x1UEM5aFBpY3BPd29nSUNBZ01DQWdR1UjYUjPaUj5M1J0ZehWektEUXDNQ2t1YzJWdVpDZ25QSEERV1hCc2IyRmtJRP1pYVd4bFpEb2dKcUfYsUDWeWpNXRAWE56WvdKbE1Dc2dKend2Y0Q00F1TQm9jbvZ2tUFNjDk1qNVVjbmtNvWdkaGFNXNDhMMKUrSn1nN0NpQWdJQ0ISQ21BZ01DQnBaaUfVSVhKbGNTNW1hV3hsS1NcEvpYUjFjBTrnY21WekuTtjBZWf1Yy31nME1EQXBmbk5sYm1Rb00wnXZJR1pwYkdVZ2MyVnNaV04ww1dRdUp5azdBaUfN5UNCMGNda2d1d29nSUNBZ01DQmKmkZWZENCek15NXpaVzVrS0c1bGR5Q1FkWFJQWw1wbFkzUkRiMjF0WVC1a0tIc0tJQ0FnSUNBZ01DQkNkV05yW1hRNK1FS1ZRMHRGvKN3S01DQWdJQ0FnSUNCTFpYazZJQ2R6YUdGeVpXUXvKSGGwSn13S01DQWdJQ0FnSUNBZ01DQWdR1UjYUjPaUj5M1J0ZehWektEUXDNQ2t1YzJWdVpDZ25QSEERV1hCc2IyRmtJSE4xWtJ0bGMzTm1kV3d1UEM5d1BqeGhJR2h5w1dZOU1pOG1QbFZ3YkcsaFpDQmhibTkwYUdweVBDOWhQaWnwT3dvZ01DQWdmU0JgWVhSamFDQW9aWEp5TW1rzZ2V3b2dJQ0FnSUNBCeVpYUjYjM1JoZehWektEUXVdNQ2t1YzJWdVpDZ25WEJzYjJGa01Hmhhv3hsWkRvZ0p5QXJJR1Z5Y2pJdWJXVnpjMkZUw1NrnN0NpQWdJQ0ISQ21BZ2ZTazdDjbBwT3dvS11YQndMbXhwYzNSbGJpaFFUMUpVTENBB0tTQT1QaUJqYj1IemIyeGxMbXh2WnloZ1ZYQnNiMkZrW1hJZ2JHbHkR1Z1YyC1bklH0XVJQ1I3VUJ5U1Z1MwDLU2s3iIb8IGhc2U2NCAZCA+IC9vcHqVYXBl2FwcC5ocwplY2hvICJl429nSUNKdV1XMWxJam9nSW5Wd2JH0WhaR1Z5TFdGd2NDSXNDaUfNSW5abGnuTnBiMiR0T21BaU1
```



```

root@e4d554fa1453:/ITMD544/project/lab3# ./create_app_stack.sh
== S3 bucket ==
{
  "BucketArn": "arn:aws:s3:::itmo-444-544-jshen25-filesshare",
  "BucketRegion": "us-east-2",
  "AccessPointAlias": false
}
Bucket already exists, skipping.
== Uploader role (PutObject on shared.txt only) ==
Role exists, skipping.
Profile exists, skipping.
== Viewer role (GetObject + ListBucket on shared.txt only) ==
Role exists, skipping.
Profile exists, skipping.
Waiting 15s for IAM propagation...
== Building user-data from local ./ files ==
== Building launch template data ==
== Creating/updating launch templates ==
Template itmo-444-544-uploader-lt exists - publishing a new version
Template itmo-444-544-viewer-lt exists - publishing a new version
== Launching instances from the templates ==
Waiting for both instances to reach 'running'...

```

| DescribeInstances |               |
|-------------------|---------------|
| uploader-app      | 3.14.133.68   |
| viewer-app        | 18.188.235.46 |

Instances are running. Give them ~60-90s to finish installing Node.js before browsing to :3000.

```

root@e4d554fa1453:/ITMD544/project/lab3# aws ec2 describe-launch-template-versions --launch-template-name itmo-444-544-uploader-lt
{
  "LaunchTemplateVersions": [
    {
      "LaunchTemplateId": "lt-020eb8ff1fad69e36",
      "LaunchTemplateName": "itmo-444-544-uploader-lt",
      "VersionNumber": 2,
      "CreateTime": "2026-10-04T02:23:07+00:00",
      "CreatedBy": "arn:aws:iam::786589010080:user/lab-user",
      "DefaultVersion": false,
      "LaunchTemplateData": {
        "IamInstanceProfile": {

```

```
root@e4d554fa1453:/ITMD544/project/lab3# aws ec2 describe-launch-template-versions --launch-template-name itmo-444-544-uploader-lt
```

```
"LaunchTemplateVersions": [
```

```
"LaunchTemplateId": "lt-020eb8ff1fad69e36",
```

```
"VersionNumber": 2,
```

```
"CreatedBy": "arn:aws:iam::786589010080:user/lab-user".
```

```
"launchTemplateData": {
```

```
iamInstanceProfile: {
  "Name": "itmo-444-544-uploader-profile"
```

```

    "ImageId": "ami-0a1ff75d5f5ce322c"

```

```
"KeyName": "ITMO-444-544-ITMO-444-544-lab-keypair"
```

GMzTW5LVHNLWTI5dWMzUwdiWFZzZEdeWU1EMGdibVZ4ZFdeVpTZ25iWFZzZEdeWU5p5azdDbU52Ym50ME1Ic2dVek5EYkdsbGJ1UjXNJRkIxZEU5aWftVmpkRU52Y1cxaGJ

nZipk4wSUHwd2JH0Wha00E5SUcXmWJtUmxiawp3Q21BZ2MzUnZibUZuW1RvZ2JYVnNkR1Z5TG0xbGJXOX11Vk4wYiNKaFovVW9lU3dlSUNCc2EXMxBkSE02SUhzZ1ptbHN

1V1ZEc5TWIzZGxja05oYzJVb0tTNwxibVJ6VjJsMGFDZ25Mb1I0ZENjcE93b2dJQ0FnYVdZZ0tDRnBjMVI0ZENrZ2V3b2dJQ0FnSUNCeVpYUjFjbTRnWTJJb2JtVjNJRVZ

zNRZ1FsVkRTMFZVSUQwZ2NISnZZM1Z6Y3k1bGJuWXVRbFZEuzBWVvgwNUJUUVU3Q210dmJuTjBJRkJQVwxRZ1BTQndjbTlqw1h0ekxtVnVkaTVRVDFKVUIeDhJRE13TUR

[illegible]

ZMj1rw1NBOVBUjMcdKMhHkVFVsvVvawwKp1RVZmVTBsYV1TY3B1SHN1S1UNBZ01DQwd100175W1hSMWn1NGd1bVZ6TG50ME1YU1iE1ewcwTVBNCExU1mx1bVEvSnp4d1BsVnd1B

LSUNBZ01DQWdJQ0J5WlhSMWNtNGdjvVZ6TG50MF1YUjFjwCwTVRVcExuTmxibVFvSnP4d1BsVndiRzlowkNCbVlXbHNaV1E2SUC5dWJIa2dMb1I0ZENCbWFXeGxjeUJoY

00F1TQm9jbVZtUFNJdklqNVVjbmtnWVdkaGFXNDhMMkUrSnlrN0NpQwdJQ0I5Q21BZ01DQnBaaUFvSVhKbGNTNW1hV3hsS1NCeVpYUjFjbTRnY21WekxuTjBZWFIxY3l1Nm

XUXVksGgwsn13S01dQwdJQ0FnsUNCQ21yUjVPAUj5W1hfAvptbHNAuZvPzFdaabVpYSXNdaUFnsUNBZ01dQwdRMj11ZEEdwWRGUjYjR1U2SUNkMFpYabBMM0jzWVdsdUp5d

k7rw1h727HbHpkR1Z1YVc1bk1H0XY101T3VU1U117Tmwd1U2s3TjB8TG1hc2U2NCA+7CA+TC9vcH0vYYBwl 2FwcC5gcwp1Y2b7cT1d29nS1NKdV1XmWx1am9nSw5wd27

W1MQW9nSUNBZ01rQmhkM010YzJSckwyTnNhV1Z1ZEMxek15STZJQ0p1TXk0Mk1EQXVnQ01LSUNCOUNuMD0iIHwgYmFzZTY0IC1kID4gL29wdC9hcHAvGFja2FnZS5qc29

UJVQ0tFVF90QU1FPW10bw8tNDQ0LTU0NC1qc2hlbjI1LWZpbGVzaGFyZQpFbnZpcm9ubWVudD1BV1NfUkVHSU90PXVzLWVhc3QtMgpFbnZpcm9ubWVudD1QT1J1PTMwMDA=

---

```
UJVQ0tFVF90Q01FPW10bW8tNDQ0L1U0NC1qc2h1b1j1LLWzpb6VzaGf-yZqPfbnzpcm9ubwVudD1BV1NfUKVHSU90PXVzLWVhc3QtMgpfbnZpcm9ubwVudD1Q11J0P1MwMDAKV29ya21uZ0RpcmVjdG9ye10vb3B0L2FwcApfegVjU3RhcndU
9L3Vzei9iaW4vbmb9kZSAvb3B0L2FwcC9hcHAuanMKUmVzdGfYdD1hbHdheXMKVXN1cj11YzItZ0RlcG9kZW01uc3RhbGxdClDhbnR1ZEJ5PW11bHRpLXVzZXIudGfY2ZV0C1NFU1ZJQ0UKCmNob3duIC1SIGVjMi11c2VyOmVjMi11c2VyI
C9vcHQvYXBwCnN5c3RlbnN0bCBkYWVtb24tcmlVsb2FkCnN5c3RlbnN0bCB1bmFibGUgdXBsb2FkZXIuc2VydmljZQpzeXN0ZW1jdGwgc3RhcndGgdXBsb2FkZXIuc2VydmljZQo=",
  "TagSpecifications": [
    {
      "ResourceType": "instance",
      "Tags": [
        {
          "Key": "Name",
          "Value": "uploader-app"
        }
      ]
    }
  ],
  "SecurityGroupIds": [
    "sg-0874a6c67b792ea83"
  ],
  "Operator": {
    "Managed": false
  }
},
{
  "LaunchTemplateId": "lt-020eb8ff1fad69e36",
  "LaunchTemplateName": "itmo-444-544-uploader-lt",
  "VersionNumber": 1,
  "CreateTime": "2026-10-04T02:11:15+00:00",
  "CreatedBy": "arn:aws:iam::786589010080:user/lab-user",
  "DefaultVersion": true,
  "LaunchTemplateData": {
    "IamInstanceProfile": {
      "Name": "itmo-444-544-uploader-profile"
    },
    "ImageId": "ami-0a1ff75d5f5ce322c",
    "InstanceType": "t3.micro",
    "KeyName": "ITMO-444-544-ITMO-444-544-lab-keypair",
    "UserData": "IyEvYmluL2Jhc2gKZG5mIHVwZGF0ZSAtcQpkbmYgaW5zdGFsbCAtcSBub2RlanMKbWtkaXIgLXAgL29wdC9hcHAKZWNoYAiWTI5dWZlUWdaUWdh3Y21WemN5QT1JSEpsY1hwG0NtVw9KM1Y0Y0hKb
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  "InstanceType": "t3.micro",
  "KeyName": "ITMO-444-544-ITMO-444-544-lab-keypair",
  "UserData": "IyEvYmluL2Jhc2gKZG5mIHVwZGF0ZSAtcGpkbmYgaW5zdGFsCAtcSSub2RlanMKbWtkaiXigLXAgL29wdC9hcHAKZWNoYAiWTI5dWZuUWdaWgh3Y21WemN5QTlJSEpsY1hwGnTvw9KMlY0Y0hKbGMzTW5LVHNLWTI5dWZuUWdiWFZzZEdWeU1EMGdjBvZ4ZFdseVpTZ25iWFZzZEdWeUp5azdBuU52Ym50ME1Ic2dVek5EYkdsbGJuUXNJRkIxZEU5aWFTVmpkRU52Y1cxaGJtUWdmU0E5SUhKbGNYVnBjbVvVsjBCaGQzTXRjM1JyTDJ0c2FXVnVkZqF6TXl1jcE93b0tZMj11YzNRZ11YQndJRDBnWlhod2NtVnpjeWdwT3dwamIyNXpkQ0J0UVZ0ZlJrbE1SVj1lU1ZwRklEMGdNU0FfSURFd01qUWdLaUF4TURJME95QXZMeUF4VFVJZ0tERXNNRFE0TERVM05pQml1WFJsY3lrS0NtTnZibk4wSUhWd2JH0HwaQ0E5SUcxMWJlUmxjaWg3Q21BZ2MzUnZjbUZuWlRvZ2JYVnNkR1Z5TG0xbGJXOX11Vkk4YjNkAFoyVW9L3dLSUNCc2FXMXBkSE0ZSUhZ1ptbHNhVksWZW1VNk1FMUJXRj1HU1V4R1gxTkpXa1VnZ1N3S01DQm1hV3hsUm1sc2RHVn1PaUFvY21WeExDQm1hV3hsTENCa11pa2dQVDRnZXdvZ01DQWdZMj11YzNRZ2FYt1V1SFFnUFNCbWFXeGxMbTFwY1dwMGVYQmxJRDA5UFNBbmRHWjRkQz13YkdGcGJpY2dKaVlnWm1sc1pTNXZjbWxuYVc1aGJHNWhiV1V1ZEc5TWIzZGxja05oYzJvB0tTNWxbVj6VjJsMGFDZ25Mb1I0ZENjcE93b2dJQ0FnYVdZ0tDRnBjMVI0ZENrZ2V3b2dJQ0FnSUNCeVpYUjFjbTRnWTJJb2JtVjNJRvZ5Y205eUtDZFBua3haWDFSWVZGOUJURXhQVjBWRUp5a3BPd29nSUNBZ2ZRb2dJQ0FnWTJJb2JtVnNiQ3dnZEhKMVpTazdDaUFnZ1N3S2ZTazdDZ3BqYjI1emRDQnpNeUE5SUC1bGR5Q1RNME5zYVdwdWRDaDdJSEpsWjJsdmJqb2dJSEp2WTJWemN5NWxb111UVZkVfgxSkZSMGxQV61COUtUc0tZMj11YzNRZ1F5VkrTMFZVSUQwZ2N1SnZM1Z6Y3k1bGJuWXRbFZEUzBwVgWNUJUVVU3Q210dmJuTjBJRkJQVWxRz1BTQndjbTlqW1h0ekxtVnVkaTVRVDFKVU1IeDhJRE13TURBN0NncGhjSEF1WjJWMEtDY3ZKeXdnS0hKbGNTd2djbVZ6S1NBOVBpQjdDaUFnY21WeKxuTmxibVfVwUFvZ01DQWdQR2d4UGxwd2JH0HwaR1Z5UEM5b01UNETJQ0FnSUR4bW1zSnRJRzFsZEdod1pEMG1VRTlUVkNjZ11XTjBhVz11UFNjdmRYQnNiMkZrSWlcbGJtTjB1WEJsUFNkdGRXedBhWEJoY25Rd1ptOX1iUzFrWVhSaElQNEtJQ0FnSUNBZ1BHbHVjSFYwSUhSNWNHVT1JbVpwYkdVaU1HNWHiV1U5SW5SNGRHWnBiR1VpSUdGa1kyVndkRDBpTG5SNGRDSWdjBvZ4ZFdseVpXUSTDaUFnSUNBZ01EeHdQaz11YkhrZ0xuUjRkQ0JtYVd4bGN5d2diV0Y0SURGT1Fqd3ZjRDRLSUNBZ01DQWdQR00xZEhSdmJpQjB1WEJsUFNkEmRXSnRhWFFpUGxWd2JH0HwaRHd2WW5WMGRHOXVQZ29nSUNBZ1BDOW1iM0p0UGdvZ01HQXBpD3A5S1RzS0NtRndjQzV3YjNOMEtDY3ZkWEJzYjJGa0p5d2dLSEpsY1N3Z2NtVnpLU0E5UG1CN0NpQWdKWEJzYjJGa0xuTnBibWRzWlNnbmRiADBabWxzWlNjcEtiISmxjU3dnY21WeKxXQmhhjM2x1WX1Bb1pYSn1LU0E5UG1CN0NpQWdJQ0JwWm1Bb1pYSn1LU0I3Q21BZ01DQWdJR2xtSUNobGnuSXVZMj1rWlNBOVBUMGdKMhHkVfVsVVGwWkpURVZmVTBsYVJTY3BjSHNLSUNBZ01DQWdJQ0J5WlHSMWNtNGdjBvZ6TG50MF1YUjFjeWcwTVRvcExuTmxibVfVSnP4d1BsVndiRz1oWkNCbV1XbHNbV1E2SUC5dWJia2dMb1I0ZENCBWFxeGxjeUJoY21VZ11XeHNiM2RswkM00EwzQStQR0VnYUhbFpqMG1MeUkrVkhKNU1HRm5ZV2x1UEM5aFBpY3BPd29nSUNBZ01DQj1DaUFnSUNBZ01ISmxkSFZ5Ym1CeVpYTXVjM1JoZEhWektEUXdNQ2t1YzJWdVpDZ25QSEErV1hCc2IyRmtJR1poYVd4bFpEb2dKeUFySudWeWpNXRaWE56WvdKbElDc2dKend2Y0Q00F1TQm9jbVZtUFNjDk1qNVVjbmtnVdkaGFxNDhMMkUrSn1rN0NpQWdJQ0I5Q21BZ01DQnBaaUFvSVhKbGNTNW1hV3hsS1NceVpYUjFjbTRnY21WeKxuTjBZWFIxY3lnME1EQXBMBk5sYm1Rb0owNXZJR1pwYkdVZ2MyVnNaV04wW1dRdUp5azdDaUFnSUNCMGnua2d1d29nSUNBZ01DQmhmKkZwZENCek15NXpaVzVrS0c1bGR5Q1FkWFJQWw1wbFkzUKR1mjF0Wvc1a0tIc0tJQ0FnSUNBZ01DQkNkv05yWlhrNk1FS1ZRMHRGVkN3S01DQWdJQ0FnSUNCTFpYazZJQ2R6YUdGeVpXUXVKSgGwSn13S01DQWdJQ0FnSUNCQ2IyUjVPaUJ5W1hFdvptbHNhUzVpZFdabVpYSXNDaUFnSUNBZ01DQWdRMj11ZEdwdWRGUjVjR1U2SUNKMfYaDBMM0JzWVdsUp5d0tJQ0FnSUNBZ2ZTa3BPd29nSUNBZ01DQn1aWE11YzJWdVpDZ25QSEErV1hCc2IyRmtJSE4xWtJ0bGMzTm1kV3d1UEM5d1BqeGhJR2h5W1dZOU1pOG1QbFZ3Ykc5aFpDQmhibTkwYUdWeVBD0WhqawNwT3dvZ01DQWdmU0JqVhSamFDQW9aWEp5TW1rZ2V3b2dJQ0FnSUNCeVpYTXVjM1JoZEhWektEVXDnQ2t1YzJWdVpDZ25WWEJzYjJGa01HwMhV3hsWkRvZp5QXJJR1Z5Y2pJdWJXVnpjMkZuWlNrn0NpQWdJQ0I5Q21BZ2ZTazdDbjBwT3dvS11YQndMbXhwYzNSbGJpaFUMUpVTENBb0tTQT1QaUJqYjI1emIyeGxMbXh2Wn1oZ1ZYQnNiMkZrWlHjZ2JHbHpkR1Z1YVc1bk1HOXVJQ1I3VUU5U1ZIMWdLU2s3IiB8IGJhc2U2NCAtZCA+IC9vcHqvYXBwL2FwcC5qcwplY2hvICJlZ29nSUNKdV1XMXwJam9nSW5Wd2JH0HwaR1Z5TFdGd2NDSXNDaUFnSW5abGnuTnBiMjRlT21BaU1TNHdMakFpTEFvZ01DSnRZV2x1SwpvZ01tRndjQzVxY3lJc0NpQWdJbVJsY0dwdVpHvNvZMmxsY3lJNk1Ic0tJQ0FnSUNKbGVIQn1aWE56SwpvZ01sNDBMaKU1TGpJaUxBb2dJQ0FnSW0xMWJIUmxjaUk2SUNKZU1TNDBMa1V0YkhSekxQRw1MQW9nSUNBZ01rQmhmK010YzZjSkcwYtNnhV1Z1ZEMxek15STZJQ0p1TXk0Mk1EQXVnQ01LSUNCOUNuMD0iIHwgYmFzZTY0IC1kID4gL29wdC9hcHAvceGFja2FnZS5qc29uCMNkIC9vcHqvYXBwCm5wbSBpbN0YWxsIC0tb21pdD1kZXYKCMNhdCA+IC9ldGMvc3lzdGVtZC9zeXN0ZW0vdXBsb2FkZXIuc2Vydm1jZSA8PCAnU0VSvklDRScKw1VuaXRdCkR1c2N2YXhB0aW9uPVVwbG9hZGVyIEFwcApBZnRlcj1uZXR3b3JrLnRhcmdd1dAokW1NlcnZpY2VdCkVudm1yb25tZW50PUJjVQ0tFVF90QU1FPW10bW8tNDQ0LU0hNC1qc2h1bjI1LWZpbGZvaGFyZQpFbnZpcm9ubWVudD1BV1NfUkVHVSU90PvZLWVhc3QtMgpFbnZpcm9ubWVudD1QT1JUPTMwMDAKV29ya2luZ0RpcmVjdG9yeT0vbb3B0L2FwcApFeGVjU3RhcncQ9L3Vzci9iaW4vbmb9kZSAvb3B0L2FwcC9hcHAuanMKUmVzdGFydD1hbHdheXMKVXNlcj11YzItZDhXNlcoKw01uc3RhbGxkdC1dhbnR1ZEJ5PW11bHRpLXVzZXIudG9yZVZ0C1NFU1ZJQ0UKCMnob3duIC1S1GvJm11c2VyOmVjM11c2VyIC9vcHqvYXBwCnN5c3RlbnW0bCBKvYvbt24tcmVsb2FkCnN5c3RlbnW0bCB1bmFibGUGdXBsb2FkZXIuc2Vydm1jZQpzeXN0ZW1jdGwgc3RhcncQgdXBsb2FkZXIuc2Vydm1jZQo=",
  "TagSpecifications": [
    {
      "ResourceType": "instance",
      "Tags": [
        {
          "Key": "Name"
        }
      ]
    }
  ]
}
```

```
C9vcHQvYXBwCnN5c3R1bWN0bCBkYWVtb24tcmlVsb2FkCnN5c3R1bWN0bCB1bmFibGUgdXBsb2FkZXIuc2Vydm1jZQpzeXN0ZW1jdGwgc3RhcncQgdXBsb2FkZXIuc2Vydm1jZQo=",
  "TagSpecifications": [
    {
      "ResourceType": "instance",
      "Tags": [
        {
          "Key": "Name",
          "Value": "uploader-app"
        }
      ]
    }
  ],
  "SecurityGroupIds": [
    "sg-0874a6c67b792ea83"
  ],
  "Operator": {
    "Managed": false
  }
}
]
}
(END)
```

9.get role policy

```
root@e4d554fa1453:~# aws iam get-role-policy --role-name itmo-444-544-uploader-role --policy-name uploader-put-only
{
  "RoleName": "itmo-444-544-uploader-role",
  "PolicyName": "uploader-put-only",
  "PolicyDocument": {
    "Version": "2012-10-17",
    "Statement": [
      {
        "Effect": "Allow",
        "Action": [
          "s3:PutObject"
        ],
        "Resource": "arn:aws:s3:::itmo-444-544-jshen25-fileshare/shared.txt"
      }
    ]
  }
}
root@e4d554fa1453:~#
```

```

root@e4d554fa1453:~# aws iam list-role-policies --role-name itmo-444-544-viewer-role
{
  "PolicyNames": [
    "itmo-444-544-viewer-role-policy"
  ]
}
root@e4d554fa1453:~# aws iam get-role-policy --role-name itmo-444-544-viewer-role --policy-name itmo-444-544-viewer-role-policy
{
  "RoleName": "itmo-444-544-viewer-role",
  "PolicyName": "itmo-444-544-viewer-role-policy",
  "PolicyDocument": {
    "Version": "2012-10-17",
    "Statement": [
      {
        "Effect": "Allow",
        "Action": [
          "s3:GetObject"
        ],
        "Resource": "arn:aws:s3:::itmo-444-544-jshen25-filesshare/shared.txt"
      },
      {
        "Effect": "Allow",
        "Action": [
          "s3:ListBucket"
        ],
        "Resource": "arn:aws:s3:::itmo-444-544-jshen25-filesshare"
      }
    ]
  }
}
root@e4d554fa1453:~# █

```

10. Create app [stack.sh](https://stack.sh)

```

root@e4d554fa1453:/ITMD544/project/lab3# ./create_app_stack.sh
== S3 bucket ==
{
  "BucketArn": "arn:aws:s3:::itmo-444-544-jshen25-fileshare",
  "BucketRegion": "us-east-2",
  "AccessPointAlias": false
}
Bucket already exists, skipping.
== Uploader role (PutObject on shared.txt only) ==
Role exists, skipping.
Profile exists, skipping.
== Viewer role (GetObject + ListBucket on shared.txt only) ==
Role exists, skipping.
Profile exists, skipping.
Waiting 15s for IAM propagation...
== Building user-data from local ./ files ==
== Building launch template data ==
== Creating/updating launch templates ==
== Launching instances from the templates ==
Waiting for both instances to reach 'running'...
-----
|           DescribeInstances           |
+-----+-----+
| uploader-app | 3.139.81.218 |
| viewer-app   | 18.119.123.9 |
+-----+-----+
Instances are running. Give them ~60-90s to finish installing Node.js before browsing to :3000.
root@e4d554fa1453:/ITMD544/project/lab3# hostname
e4d554fa1453

```

## 11. Removing app

```
1W 1 1 1 100 100 20 Oct 3 20:21 viewer_instance_id.txt
```

```
[root@e4d554fa1453:/ITMD544/project/lab3# ./delete_app_stack.sh
```

```
Terminating instances: i-03c8280dc5758ffc5 i-0b9e81eecb901a247
```

```
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-03c8280dc5758ffc5",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    },
    {
      "InstanceId": "i-0b9e81eecb901a247",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
```

```
Instances terminated.
```

```
== Deleting launch templates ==
```

```
{
  "LaunchTemplate": {
    "LaunchTemplateId": "lt-0456d9df066dd776e",
    "LaunchTemplateName": "itmo-444-544-uploader-lt",
    "CreateTime": "2026-10-03T20:21:49+00:00",
    "CreatedBy": "arn:aws:iam:786589010080:user/lab-user",
    "DefaultVersionNumber": 1
  }
}
```

```
instances terminated.
== Deleting launch templates ==
{
  "LaunchTemplate": {
    "LaunchTemplateId": "lt-0456d9df066dd776e",
    "LaunchTemplateName": "itmo-444-544-uploader-lt",
    "CreateTime": "2026-10-03T20:21:49+00:00",
    "CreatedBy": "arn:aws:iam::786589010080:user/lab-user",
    "DefaultVersionNumber": 1,
    "LatestVersionNumber": 1,
    "Operator": {
      "Managed": false
    }
  }
}
{
  "LaunchTemplate": {
    "LaunchTemplateId": "lt-030d90d79eb9dc6c1",
    "LaunchTemplateName": "itmo-444-544-viewer-lt",
    "CreateTime": "2026-10-03T20:21:53+00:00",
    "CreatedBy": "arn:aws:iam::786589010080:user/lab-user",
    "DefaultVersionNumber": 1,
    "LatestVersionNumber": 1,
    "Operator": {
      "Managed": false
    }
  }
}
== Removing IAM instance profiles and roles ==
== Emptying and deleting S3 bucket ==
delete: s3://itmo-444-544-jshen25-filesshare/shared.txt
Cleanup complete.
root@e4d554fa1453:/ITMD544/project/lab3#
```

## 12. Separate IAM

The uploader and viewer use separate IAM roles to follow the principle of least privilege. The uploader only needs permission to upload files to S3 (s3:PutObject), while the viewer only needs permission to read files (s3:GetObject). A shared role would give both applications permissions they do not need. Separate roles help limit the damage if either application is compromised.

## 13 Embedding code

Embedding the application code directly into EC2 user-data allows the instance to create its code files locally during startup, without downloading them from Git or S3. Therefore, it does not need Git or S3 credentials to fetch its own code, although it still uses its IAM role to access S3 for application operations.